SUMS OF TWO SQUARES WITH FEW PRIME FACTORS VIA THETA SERIES OF QUATERNARY FORMS

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ABSTRACT. Let $N \equiv 2 \pmod 4$ be large. We show that the number of $n \le N$ such that n and $N-n$ are both sums of two squares, $\Omega(n(N-n)) \le 9$ and $\max(\Omega(n), \Omega(N-n)) \le 6$ is $\gg N/(\log N)^2$. The previous bounds, obtained by sifting the primes $\equiv 3 \pmod 4$ and switching, were 11 and 8. The new input is a level of distribution $2/5$ for the weighted sequence $r(n)r(N-n)$, where r counts representations as a sum of two squares: for coprime m, m' the sum $\sum_{ma+m'b=N} r(a)r(b)$ is the N-th coefficient of the weight-two theta series of $m(x^2+y^2)+m'(z^2+w^2)$, and Waibel's uniform bound for the Petersson norm of the cuspidal part of a theta series, combined with Deligne's bound through the Hecke operators, gives $\sum_{ma+m'b=N} r(a)r(b) = (\text{Siegel main term}) + O(d(N)N^{1/2}(mm')^{1/4+\varepsilon})$ uniformly in m, m'. Numerically the error does not grow with mm' at all, which suggests that the level of distribution is in fact $1/2$.

1. INTRODUCTION

Let $r(n) = \#\{(x,y) \in \mathbb{Z}^2 : \ x^2+y^2 = n\}$, so that $r(n) = 4\sum_{d|n} \chi(d)$ with $\chi = \chi_{-4}$ the non-trivial character modulo 4. Every integer N is a sum $n + (N-n)$ of two sums of two squares, and in [3] we asked for such representations in which n and $N-n$ have few prime factors. For $N \equiv 2 \pmod 4$ it was shown there, by a vector sieve in which the primes $q \equiv 3 \pmod 4$ are sifted up to nearly $N^{1/2}$ and the survivors with two large such primes are removed by a switching argument, that there are $\gg N/(\log N)^2$ integers $n \le N$ with n, $N-n$ sums of two squares and $\Omega(n(N-n)) \le 11$, and $\gg N/(\log N)^2$ with $\max(\Omega(n), \Omega(N-n)) \le 8$.

Here we take a different route: instead of sifting the primes $\equiv 3 \pmod 4$ we count with the weight $r(n)r(N-n)$, which vanishes unless both n and $N-n$ are sums of two squares. The price is that the sequence $r(n)r(N-n)$ must be shown to be well distributed in arithmetic progressions, and the gain is that no inert prime has to be sifted and no switching is needed. We prove a level of distribution $2/5$ (Theorem 3.2), and the sieve then gives the following.

Theorem 1.1. *Let $N \equiv 2 \pmod 4$ be sufficiently large. Then*

$$\#\big\{n \le N : \ n = x^2+y^2, \ N-n = z^2+w^2, \ \Omega(n(N-n)) \le 9, \ \max(\Omega(n), \Omega(N-n)) \le 6\big\} \ge \ c_1 \, \mathfrak{S}_r(N) \, \frac{N}{(\log N)^2},$$

where $c_1 > 0$ is an absolute constant and

$$\mathfrak{S}_r(N) = \prod_{\substack{p|N \\ p>2}} \rho_p, \qquad \rho_p = \begin{cases} \dfrac{p(p+1)(p-1)^2}{(p^2-p+1)^2} & p \equiv 1 \pmod 4, \\ \dfrac{p+1}{p}\Big(1 - \dfrac{1}{p(p-1)}\Big)^{-2} & p \equiv 3 \pmod 4. \end{cases} \tag{1}$$

One has $\rho_p > 1$ for every p, so $\mathfrak{S}_r(N) \ge 1$. The integers counted are coprime to N, and each of n, $N-n$ is a product of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{9/250}$.

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The constant can be made explicit, $c_1 = 0.25$ being admissible (Section 4); we have not tried to optimise it. Both bounds 9 and 6 come from one set of sieve parameters, so they hold simultaneously for the same integers n.

The method. Write a square-free d as a product of primes $p \equiv 1 \pmod 4$ (split) and $q \equiv 3 \pmod 4$ (inert). The identities $r(n) = 2r(n/p) - r(n/p^2)$ for split $p \mid n$ and $r(n) = r(n/q^2)$ for inert $q \mid n$ express the sum of $r(n)r(N-n)$ over $n \equiv 0\ (d)$, $n \equiv N\ (e)$ as a combination of the sums

$$T(m,m';N) = \sum_{ma+m'b=N} r(a)r(b) = \#\{(x,y,z,w) \in \mathbb{Z}^4 :\ m(x^2+y^2)+m'(z^2+w^2) = N\} \tag{2}$$

with $m \mid d^2$, $m' \mid e^2$. This is the number of representations of N by the positive quaternary form $Q_{m,m'} = m(x^2+y^2)+m'(z^2+w^2)$, i.e. the N-th Fourier coefficient of the theta series $\theta^2(mz)\theta^2(m'z)$, a holomorphic modular form of weight 2 on $\Gamma_0(4mm')$. By Siegel's theorem it is the sum of the N-th coefficient of an Eisenstein series, which is the product $M(m,m';N)$ of the local densities, and of the N-th coefficient of a cusp form $G_{m,m'}$. Deligne's bound controls the latter for each fixed form, and the whole difficulty is the uniformity in the level $4mm'$. This is exactly what Waibel's recent theorem on the Petersson norm of the cuspidal part of a theta series provides [12]: for our forms it gives $\|G_{m,m'}\| \ll (mm')^{1/4+\varepsilon}$. Passing from the norm to the N-th coefficient through the Hecke operator T_N rather than through the Petersson formula directly avoids a loss for small levels, and we obtain

$$T(m,m';N) = M(m,m';N) + O_\varepsilon\big(d(N)^4 N^{1/2}(mm')^{1/4+\varepsilon}\big) \tag{3}$$

uniformly in coprime odd cube-free m, m' (Theorem 2.3 and Corollary 2.6; for $(N, mm') = 1$ the factor $d(N)^4$ can be replaced by $d(N)$). Since the main term is of size $N/(mm')$, the error is smaller than the main term as long as $mm' \le N^{2/5-\varepsilon}$, and a short bookkeeping argument turns (3) into the level of distribution $2/5$ for $r(n)r(N-n)$ (Theorem 3.2).

The weighted sequence has the pleasant feature that under the weight $r(n)r(N-n)$ a split prime p divides n with density $h(p) = (2p-1)/(p(p+1)) \approx 2/p$ and an inert prime only with density $1/(q(q-1))$; so sifting all primes below N^α on both sides is a vector sieve of two linear components, and Richert's weights bound $\Omega(n(N-n))$. The numerical inequality behind Theorem 1.1 is verified with interval arithmetic (Section 4); it is considerably simpler than the one in [3], because there is no switching term.

Comparison with the shifted convolution literature. The sum (2) is a shifted convolution problem for the coefficients of θ^2 with linear coefficients m, m', in the "sum" orientation $ma+m'b = N$. For the divisor function in place of r and the "difference" orientation, Topacogullari [11] proved an asymptotic formula uniform in the coefficients with error $(r_2x_1)^{1/2+\vartheta+\varepsilon}$, where $\vartheta = 7/64$ is the Kim–Sarnak exponent; the analogue of his result for $r(n)$ in our orientation would give the level $1/2 - \vartheta = 25/64$. The holomorphic structure of our problem does better in two ways: the sum orientation makes $T(m,m';N)$ a coefficient of a holomorphic form, so that Deligne's theorem replaces the Ramanujan–Petersson bound at finite places, and the uniformity in the level is supplied by the Petersson norm. The exponent $1/4$ in (3) is the price of a Cauchy–Schwarz step in the passage from the norm of $G_{m,m'}$ to a single coefficient. Numerically (Section 5) the quantity $\sup_N |T-M|/\sqrt{N}$ does not grow with mm', while the root mean square decreases like $(mm')^{-1/4}$, which shows that Waibel's bound is sharp up to N^ε but suggests that the error in (3) is really $O(N^{1/2+\varepsilon})$ uniformly. If so, the level of distribution is $1/2$ and the bounds in Theorem 1.1 improve to 8 and 5 (Remark 4.5).

Notation. p denotes a prime $\equiv 1 \pmod 4$ and q a prime $\equiv 3 \pmod 4$ when the distinction matters; $\chi = \chi_{-4}$; $d(n)$ and $\omega(n)$, $\Omega(n)$ are the usual divisor and prime-factor counting functions; $\sigma_{-1}(n) = \sum_{t|n} 1/t$. The Petersson inner product on $S_2(\Gamma_0(L))$ is

$$\langle f, g\rangle = \int_{\Gamma_0(L)\backslash\mathbb{H}} f(z)\overline{g(z)}\,\mathrm{d}x\,\mathrm{d}y, \qquad \|f\|^2 = \langle f, f\rangle, \tag{4}$$

which is [6, (2.1)] and the normalisation of [12] for weight 2. All implied constants may depend on ε.

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2. Theta series of the forms $m(x^2+y^2)+m'(z^2+w^2)$

Throughout this section m, m' are coprime odd positive integers, $Q = Q_{m,m'}$ is the quaternary form $m(x^2+y^2)+m'(z^2+w^2)$ with Gram matrix $A = \mathrm{diag}(2m, 2m, 2m', 2m')$, and

$$\Theta(z) = \theta^2(mz)\theta^2(m'z) = \sum_{n\ge 0} T(m, m'; n)e(nz), \qquad \theta(z) = \sum_{k\in\mathbb{Z}} e(k^2 z).$$

Since $\theta^2 \in M_1(\Gamma_0(4), \chi)$, we have $\Theta \in M_2(\Gamma_0(4mm'))$ with trivial character. In the terminology of [12], Q is primitive (because $(m, m') = 1$), $\det A = 16(mm')^2$, and the level of Q, the least L such that LA^{-1} is integral with even diagonal, is $L = 4mm'$.

2.1. Siegel's theorem and the local densities.

By Siegel's theorem [10] (for weight 2 see Kitaoka [9] and [12, Section 2]), $\Theta = E + G$ where E is the theta series of the genus of Q (a linear combination of Eisenstein series) and $G \in S_2(\Gamma_0(4mm'))$ is a cusp form, and the coefficients of E are given by the product of local densities [12, (1)–(2)]

$$M(m, m'; n) = \frac{(2\pi)^2 n}{\Gamma(2)\sqrt{\det A}} \prod_p \beta_p(n) = \frac{\pi^2 n}{mm'} \prod_p \beta_p(n), \tag{5}$$

where $\beta_p(n) = \lim_{a\to\infty} p^{-3a}\#\{x \in (\mathbb{Z}/p^a)^4 :\ Q(x) \equiv n\ (p^a)\}$. We record the densities that we need. For $p \nmid 2mm'$ the form is equivalent over $\mathbb{Z}_p$ to $x^2+y^2+z^2+w^2$ and

$$\beta_p(n) = (1-p^{-2})\,\sigma_{-1}(p^{v_p(n)}), \tag{6}$$

as follows from Jacobi's formula $r_4(n) = 8\sum_{t|n,\,4\nmid t} t$ (or from a direct count). For $p \mid mm'$ with $p \nmid n$ we have

$$\beta_p(n) = 1 - \frac{\chi(p)}{p}\,: \tag{7}$$

indeed, if $p \mid m$ then modulo p the equation reads $m'(z^2+w^2) \equiv n$, which has $p - \chi(p)$ solutions (z, w), none of them with $z \equiv w \equiv 0$; every such solution is non-singular and lifts to $p^{3(a-1)}$ solutions modulo p^a by Hensel's lemma, so $\beta_p(n) = p^2(p-\chi(p))/p^3$. At $p = 2$ the density $\beta_2(n)$ depends only on n modulo a bounded power of 2 and on whether m, m' are $\equiv 1$ or $3 \pmod 4$; we shall only use that $\beta_2(n) \asymp 1$ for $v_2(n) \le 1$ and that for $m = m' = 1$ one has $M(1,1;n) = r_4(n)$ exactly (there are no cusp forms of level 4). In particular, for $(n, mm') = 1$,

$$\frac{M(m, m'; n)}{M(1,1;n)} = \frac{1}{mm'} \prod_{p|mm'} \frac{1-\chi(p)/p}{1-p^{-2}} = \frac{1}{mm'} \prod_{p|mm'} g(p), \tag{8}$$

where $g(p) = p/(p+1)$ for $p \equiv 1 \pmod 4$ and $g(q) = q/(q-1)$ for $q \equiv 3 \pmod 4$.

2.2. **Waibel's bound.**

Lemma 2.1. $\|G\| \ll (mm')^{1/4+\varepsilon}$.

Proof. Theorem 1 of [12] states, for a primitive integral positive quaternary form of level L and $f = \theta(Q) - \theta(\mathrm{gen}Q)$,

$$\langle f, f\rangle \ll \frac{L^{2+\varepsilon}}{F(Q,2)} + \frac{L^{1+\varepsilon}}{(\det A)^{1/4}},$$

with the inner product (4) and a genus invariant $F(Q,s)$ defined in [12, (6)] from the Jordan decompositions of Q over $\mathbb{Z}_p$: $F(Q,s) = 2^{\min(4-2r_2,s)}\prod_{p|L} p^{\mu_p(s)}$, $\mu_p(s) = \sum_{j=1}^{v_p(L)} \min(v_p(j), s)$, where $v_p(j)$ is the number of variables whose p-adic coefficient has valuation $\geq j$ and r_2 the number of binary 2-adic blocks. For $Q_{m,m'}$ and an odd prime p with $p^a \parallel mm'$ the p-adic valuations of the coefficients are $(a, a, 0, 0)$, so $v_p(j) = 2$ for $j \leq a$ and $\mu_p(2) = 2a$; at $p = 2$ all coefficients are units, $r_2 = 0$ and $\mu_2(2) = 0$. Hence $F(Q,2) = 4(mm')^2$ (and $F(Q,4) = 16(mm')^2 = \det A$, as it must). With $L = 4mm'$ and $\det A = 16(mm')^2$ the bound becomes $\langle G, G\rangle \ll (mm')^{\varepsilon} + (mm')^{1/2+\varepsilon}$. □

2.3. **From the norm to a single coefficient.** The Petersson formula gives, for an orthonormal basis $\mathcal{B}$ of $S_2(\Gamma_0(L))$ and $n \geq 1$ (see [6, Prop. 2.1 and Cor. 2.2], which are stated for all even $k \geq 2$; we write them for the unnormalised coefficients $a_f(n)$),

$$\sum_{f\in\mathcal{B}} |a_f(n)|^2 = 4\pi n\, \Delta_L(n,n), \qquad \Delta_L(n,n) = 1 + O\Big(\frac{d(L)\, d_3(n)\, (n,L)^{1/2}\, n}{L\,(n+L)^{1/2}} \log 2n\Big) \tag{9}$$

with an absolute implied constant. Applying this directly at $n = N$ would cost a factor $(1 + N^{1/4}L^{-1/2})$ for small levels (this is [5, Cor. 14.24], used in [12, (3)]), which is too expensive for us; we route the coprime part of N through a Hecke operator instead.

Lemma 2.2. *Let $L \geq 1$, $G \in S_2(\Gamma_0(L))$, and $n = n_1 n_2$ with $(n_1, L) = 1$, $n_2 \mid L^{\infty}$ and $n_2 \leq L^{1/2}$. Then*

$$|a_G(n)| \leq C\, d(n_1)\, n^{1/2}\, \|G\|$$

with an absolute constant C.

Proof. For $(n_1, L) = 1$ the Hecke operator T_{n_1} is self-adjoint on $S_2(\Gamma_0(L))$ with respect to (4) and is diagonalised by a basis of forms $f(tz)$, f a newform of some level $L' \mid L$ and $t \mid L/L'$, with eigenvalue $a_f(n_1)$ ([8, Thm. 6.20], [1]). By Deligne's theorem $|a_f(n_1)| \leq d(n_1) n_1^{1/2}$, hence $\|T_{n_1} G\| \leq d(n_1) n_1^{1/2} \|G\|$. The n_2-th coefficient of $H = T_{n_1} G$ is $\sum_{t|(n_1,n_2)} t\, a_G(n_1 n_2/t^2) = a_G(n)$ because $(n_1, n_2) = 1$. Finally, by Cauchy–Schwarz over an orthonormal basis and (9),

$$|a_H(n_2)|^2 \leq \|H\|^2 \sum_{f\in\mathcal{B}} |a_f(n_2)|^2 = 4\pi n_2 \Delta_L(n_2, n_2) \|H\|^2,$$

and for $n_2 \leq L^{1/2}$ the error term in (9) is

$$\ll d(L)\, d_3(n_2)\, \frac{n_2^{3/2} \log(2n_2)}{L^{3/2}} \ll d(L)\, d_3(n_2)\, \frac{\log(2L)}{L^{3/4}} \ll 1,$$

so $\Delta_L(n_2, n_2) \leq C'$. □

Theorem 2.3. *Let m, m' be coprime odd positive integers and $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$. Then*

$$T(m, m'; N) = M(m, m'; N) + O_{\varepsilon}\big(d(N)\, N^{1/2} (mm')^{1/4+\varepsilon}\big).$$

Proof. $T - M = a_G(N)$ with $G \in S_2(\Gamma_0(4mm'))$. Apply Lemma 2.2 with n_1 the odd part of N and $n_2 = 2^{v_2(N)} \mid 4mm'$, then Lemma 2.1. □

Remark 2.4. For fixed m, m' the error term $O(d(N)N^{1/2})$ is Deligne's bound and cannot be improved. The point of Theorem 2.3 is the dependence on mm'; see Section 5 for what it seems to be in truth.

2.4. **Primes of mm' dividing N.** The sieve also needs $T(m, m'; N)$ when a prime of mm' divides N. Such terms reduce exactly to the coprime case. Let U_p act on q-expansions by $U_p \sum a(n)e(nz) = \sum a(pn)e(nz)$, so that $a_N(\Phi) = a_{N/p}(U_p\Phi)$ when $p \mid N$, and note that $U_p\big(\Phi(pz)\Psi(z)\big) = \Phi(z)\,(U_p\Psi)(z)$. The identities $r(pb) = 2r(b) - r(b/p)$ for split p and $r(qb) = r(b/q)$ for inert q (with $r(x) = 0$ for $x \notin \mathbb{Z}$) say that for $p \nmid m'$

$$U_p\,\theta^2(m'z) = 2\theta^2(m'z) - \theta^2(pm'z) \quad (p \equiv 1\ (4)), \qquad U_q\,\theta^2(m'z) = \theta^2(qm'z) \quad (q \equiv 3\ (4)). \tag{10}$$

Lemma 2.5. *Let p be an odd prime with $p \mid N$ and $p \mid m$, $p \nmid m'$.*

(i) *If $p \equiv 1$ (4) and $m = pm_1$ with $p \nmid m_1$, then $T(pm_1, m'; N) = 2T(m_1, m'; N/p) - T(m_1, pm'; N/p)$.*
(ii) *If $p \equiv 1$ (4) and $m = p^2m_1$, then $T(p^2m_1, m'; N) = 2T(pm_1, m'; N/p) - T(m_1, m'; N/p^2)$, where the last term is 0 if $p^2 \nmid N$.*
(iii) *If $p \equiv 3$ (4) and $m = pm_1$ with $p \nmid m_1$, then $T(pm_1, m'; N) = T(m_1, pm'; N/p)$.*
(iv) *If $p \equiv 3$ (4) and $m = p^2m_1$, then $T(p^2m_1, m'; N) = T(m_1, m'; N/p^2)$, which is 0 if $p^2 \nmid N$.*

The same holds with the roles of m and m' exchanged. Consequently, if $v_p(mm') \le 2$ for every odd prime p, then $T(m, m'; N)$ is a linear combination with integer coefficients of terms $T(\tilde m, \tilde m'; \tilde N)$ with coprime $\tilde m, \tilde m'$, $\tilde N \mid N$, $v_2(\tilde N) = v_2(N)$, $(\tilde m\tilde m', \tilde N) = 1$ and $\tilde m\tilde m' \le mm'$, the sum of the absolute values of the coefficients being at most $\prod_{p\mid(mm',N)} 4v_p(N) \le 4^{\omega(N)}d(N)$. Since the identities hold for the modular forms themselves, they hold for the Eisenstein parts M and the cusp parts separately.

Proof. In case (i), $\theta^2(mz) = \Phi(pz)$ with $\Phi = \theta^2(m_1\,\cdot)$, so

$$T(pm_1, m'; N) = a_{N/p}\big(\theta^2(m_1z)\,U_p\theta^2(m'z)\big),$$

and (10) gives the claim; (iii) is the same with the second identity. In case (ii), $\theta^2(mz) = \Phi(pz)$ with $\Phi = \theta^2(pm_1\,\cdot)$, so $T = 2T(pm_1, m'; N/p) - T(pm_1, pm'; N/p)$, and $T(pm_1, pm'; N') = T(m_1, m'; N'/p)$ (or 0 if $p \nmid N'$) because the form is p times $Q_{m_1,m'}$; (iv) is similar. For the last assertion fix p with $p \mid (mm', N)$ and iterate: when $v_p(\tilde m\tilde m') = 1$ we apply (i) or (iii), when it is 2 we apply (ii) or (iv). On the right-hand sides the two coefficients are coprime again, $v_p(\tilde N)$ has decreased by at least one, and exactly one term still contains p in its coefficients (with $v_p = 1$); the others are of the final form. The coefficient of the continuing term has absolute value at most 2 (it equals 2 in (ii), which can only be the first step, and is ±1 times the previous one in (i), (iii)), so each finished term has a coefficient of absolute value at most 4, and the finished term produced by (ii) has coefficient -1. After at most $v_p(N)$ steps either $p \nmid \tilde N$ or $p \nmid \tilde m\tilde m'$, and the sum of the absolute values of all coefficients produced is at most $1 + 4(v_p(N) - 1) + 2 < 4v_p(N)$ if the first step is (ii) and at most $2v_p(N) + 1 \le 3v_p(N)$ otherwise. Treating the primes of (mm', N) one after the other gives the bound $\prod_p 4v_p(N) \le 4^{\omega(N)}d(N)$. The Eisenstein and cuspidal parts of Θ are preserved by U_p and by $z \mapsto pz$. □

Corollary 2.6. *For coprime odd m, m' with $v_p(mm') \le 2$ for all p, and N with $v_2(N) \le 1$,*

$$T(m, m'; N) = M(m, m'; N) + O_\varepsilon\big(4^{\omega(N)}d(N)^2\,N^{1/2}(mm')^{1/4+\varepsilon}\big),$$

and $4^{\omega(N)}d(N)^2 \le d(N)^4$.

3. The level of distribution of $r(n)r(N-n)$

Let $N \equiv 2 \pmod 4$ and

$$\mathcal{A} = \{n : \ 1 \le n \le N, \ (n, N) = 1\}, \qquad w(n) = r(n)r(N-n).$$

Elements of $\mathcal{A}$ are odd, and $w(n) \neq 0$ forces $n \equiv N - n \equiv 1 \pmod 4$. For coprime square-free d, e with $(de, 2N) = 1$ put

$$S(d, e) = \sum_{\substack{n \in \mathcal{A} \\ d \mid n, \ e \mid N-n}} w(n).$$

3.1. **Decomposition into theta coefficients.** For split p and inert q the relations $r(n) = 2r(n/p) - r(n/p^2)$ $(p \mid n)$ and $r(n) = r(n/q^2)$ $(q \mid n)$ give, for square-free d,

$$\sum_{d \mid n} r(n)e(nz) = \sum_{m \mid d^2} c_d(m)\, \theta^2(mz), \tag{11}$$

where c_d is multiplicative in the sense $c_d(m) = \prod_{p \mid d} c_p(p^{v_p(m)})$ with $c_p(p) = 2$, $c_p(p^2) = -1$ for split p and $c_q(q^2) = 1$, $c_q(q) = 0$ for inert q. Every m with $c_d(m) \neq 0$ has the form $m = ds$ with $s \mid d$, and $\sum_m |c_d(m)| \le 3^{\omega(d)}$. The condition $(n, N) = 1$ is removed by Möbius inversion over the square-free divisors f of the odd part of N, and the parity of n (equivalently of $a = n/fd$) by the identity $r(2a') = r(a')$: the terms with a even in $T(m, m'; N)$ sum to $T(m, m'; N/2)$. Altogether

$$S(d, e) = \sum_{f \mid N^\flat} \mu(f) \sum_{m \mid d^2} \sum_{m' \mid e^2} c_d(m) c_e(m') \sum_{f_1 \mid f^2} c_f(f_1)\, T^{\mathrm{odd}}(f_1 m, m'; N), \tag{12}$$

with $T^{\mathrm{odd}}(\cdot\,; N) = T(\cdot\,; N) - T(\cdot\,; N/2)$, where $N^\flat$ is the product of the odd primes dividing N. Each T in (12) has coprime odd coefficients and is covered by Corollary 2.6.

3.2. **The main term.** Define $M(d, e)$ by replacing each T in (12) by its Eisenstein part M. The Siegel densities are multiplicative over the primes, and the linear identities behind (11) and the Möbius inversion hold for the numbers of solutions modulo every prime power; hence $M(d, e)/M(1, 1; N)-$ type ratios are products of conditional local densities. Concretely, let X be the Eisenstein part of $S(1, 1) = \sum_{n \in \mathcal{A}} w(n)$:

$$X = \big(r_4(N) - r_4(N/2)\big) \prod_{p \mid N^\flat} (1 - g_N(p)) = 16\, \sigma(N/2) \prod_{p \mid N^\flat} (1 - g_N(p)), \tag{13}$$

where we used that $M(1, 1; n) = r_4(n)$, Jacobi's formula ($r_4(N) = 24\sigma(N/2)$, $r_4(N/2) = 8\sigma(N/2)$ for $N \equiv 2 \pmod 4$), and where $g_N(p)$ is the proportion of the p-adic solutions of $x^2+y^2+z^2+w^2 = N$ with $p \mid x^2 + y^2$. Then

$$M(d, e) = X\, h(d)\, h(e), \qquad h(p) = \frac{2p-1}{p(p+1)} \ \ (p \equiv 1 \ (4)), \qquad h(q) = \frac{1}{q(q-1)} \ \ (q \equiv 3 \ (4)), \tag{14}$$

h multiplicative.

Lemma 3.1. *Let p be odd. (i) If $p \nmid N$, the proportion of the p-adic solutions of $x^2+y^2+z^2+w^2 = N$ with $p \mid x^2 + y^2$ is $h(p)$. (ii) If $p^a \parallel N$, $a \ge 1$, this proportion is*

$$g_N(p) = 1 - \frac{\beta_p^*}{(1 - p^{-2})\sigma_{-1}(p^a)}, \qquad \beta_p^* = \begin{cases} (1 - 1/p)^3 & p \equiv 1 \pmod 4, \\ (1 - 1/p)(1 + 1/p)^2 & p \equiv 3 \pmod 4; \end{cases}$$

in particular $g_N(p) = 4p/(p+1)^2$ for $p \equiv 1 \pmod 4$, $p \parallel N$, and $g_N(q) = 0$ for $q \equiv 3 \pmod 4$, $q \parallel N$.

Proof. (i) Modulo p, with N a unit, the equation $x^2+y^2+z^2+w^2\equiv N$ has p^3-p solutions, all non-singular. Those with $x^2+y^2\equiv 0$ number $(2p-1)(p-1)$ if $p\equiv 1 \pmod 4$ ($2p-1$ choices of (x,y) on the two lines $y\equiv \pm ix$, then $p-1$ choices of (z,w) with $z^2+w^2\equiv N$) and $1\cdot(q+1)$ if $q\equiv 3 \pmod 4$. Hensel's lemma lifts each solution to $p^{3(a-1)}$ solutions modulo p^a, so the proportion is $(2p-1)(p-1)/(p^3-p)=h(p)$, resp. $(q+1)/(q^3-q)=h(q)$. (ii) The solutions with $p\nmid x^2+y^2$ are non-singular, and their number modulo p is $\sum_{u\not\equiv 0}\#\{x^2+y^2\equiv u\}\#\{z^2+w^2\equiv -u\}=(p-1)(p-\chi(p))^2$, which gives the density β_p^*; the total density is (6). □

Note that $(1-g_N(p))\sigma_{-1}(p^a)=\beta_p^*/(1-p^{-2})$ does not depend on a; this is why $\mathfrak{S}_r(N)$ in (1) has one factor per prime. The products in (13) and (14) follow from Lemma 3.1 by multiplicativity; for (14) one can also verify directly that $\sum_{m\mid d^2} c_d(m)\,g(\operatorname{rad} m)/m=h(d)$ with g as in (8): for split p, $2g(p)/p-g(p)/p^2=\frac{p}{p+1}\big(\frac{2}{p}-\frac{1}{p^2}\big)=h(p)$, and for inert q, $g(q)/q^2=h(q)$.

3.3. **The remainder.** Put $R(d,e)=S(d,e)-M(d,e)$.

Theorem 3.2. *Let $N\equiv 2 \pmod 4$, $\varepsilon>0$ and $D=N^{2/5-\varepsilon}$. Then*

$$\sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e)| \ \ll_\varepsilon\ N^{1-\varepsilon/3}.$$

Proof. Each T in (12) has coefficients $(\tilde m,\tilde m')=(f_1m,m')$ with $\tilde m\tilde m'=f_1\,ds\,es'$, $s\mid d$, $s'\mid e$, $f_1\le f^2$; they are coprime and cube-free because d,e,f are pairwise coprime and square-free. Let $\eta=\varepsilon/10$. We use two bounds for $|T-M|$: Corollary 2.6,

$$|T(\tilde m,\tilde m';N')-M(\tilde m,\tilde m';N')|\ll d(N)^4\,N^{1/2}(\tilde m\tilde m')^{1/4+\eta}\qquad (N'\in\{N,N/2\}),$$

and the trivial bound $|T-M|\le T+M\ll N^{1+\eta}/(\tilde m\tilde m')+N^\eta$, valid because there are at most $N/(\tilde m\tilde m')+1$ pairs (a,b) with $\tilde m a+\tilde m' b=N'$ and $r(a)r(b)\ll N^\eta$, while $M\ll N^{1+\eta}/(\tilde m\tilde m')$ by (5). All divisor-type factors ($d(N)^4$, $d_3(de)$, $3^{\omega(de)}$, $d(de)$, $d(N^\flat)^{O(1)}$) are $\ll N^\eta$. The terms with $f_1ss'>N^\varepsilon$ are estimated trivially:

$$\sum_{de\le D} d_3(de)\,3^{\omega(de)}\sum_{f\mid N^\flat}3^{\omega(f)}\sum_{\substack{s\mid d,\ s'\mid e,\ f_1\mid f^2\\ f_1ss'>N^\varepsilon}}\Big(\frac{N^{1+\eta}}{f_1ss'de}+N^\eta\Big)$$
$$\ll N^{1+2\eta-\varepsilon}\sum_{de\le D}\frac{d_3(de)3^{\omega(de)}d(de)}{de}\sum_{f\mid N^\flat}\frac{9^{\omega(f)}}{f}+DN^{3\eta}\ll N^{1-\varepsilon+3\eta}+N^{2/5+3\eta},$$

and those with $f_1ss'\le N^\varepsilon$ by the uniform bound:

$$\ll d(N)^4\,N^{1/2+\eta}\sum_{de\le D}d_3(de)3^{\omega(de)}(de)^{1/4}\sum_{s\mid d,\,s'\mid e}\ \sum_{\substack{f\mid N^\flat,\ f_1\mid f^2\\ f_1ss'\le N^\varepsilon}}(f_1ss')^{1/4}$$
$$\ll N^{1/2+\varepsilon/4+3\eta}D^{5/4}=N^{1-\varepsilon+3\eta}.$$

Since $3\eta=3\varepsilon/10$, both bounds are $\ll N^{1-7\varepsilon/10}\le N^{1-\varepsilon/3}$. □

Remark 3.3. The exponent $2/5$ is $1/(2(1+\beta))$ with $\beta=1/4$ the exponent of mm' in Theorem 2.3; the terms with m close to d^2, where the level mm' is as large as $(de)^2$, are harmless because of the trivial bound. A uniform bound with $\beta=0$ (Conjecture 5.1) would give the level $1/2$.

4. The sieve

We keep the notation of Section 3. Fix $0 < \alpha < \beta_1 < 2/5$ and $\lambda > 0$, and put

$$z = N^{\alpha}, \qquad y_1 = N^{\beta_1}, \qquad P = \prod_{\substack{2<p<z \\ p\nmid N}} p.$$

For $n \in \mathcal{A}$ define Richert's weight

$$w_n = 1 - \lambda \sum_{\substack{p \mid n(N-n),\ p\equiv 1\,(4) \\ z \le p < y_1}} \Big(1 - \frac{\log p}{\log y_1}\Big), \qquad W = \sum_{\substack{n\in\mathcal{A} \\ (n(N-n),P)=1}} w_n\, w(n).$$

Lemma 4.1 (Structure)**.** *Let $n \in \mathcal{A}$ with $(n(N-n), P) = 1$ and $w(n) \neq 0$. Then every prime factor of $n(N-n)$ is $\ge z$. Unless $n(N-n)$ is divisible by the square of a prime $\ge z$, both n and $N-n$ are products of distinct primes $\equiv 1 \pmod 4$, each prime dividing exactly one of them, and $w(n) = 16 \cdot 2^{\Omega(n(N-n))}$. The total weight of the exceptional n is $\ll N^{1-\alpha+\varepsilon}$.*

Proof. $(n, N) = 1$ and $(n(N-n), P) = 1$ leave only prime factors $\ge z$. An inert prime divides a sum of two squares only to an even power, so if $n(N-n)$ is square-free all its prime factors are split, and then $r(n) = 4 \cdot 2^{\omega(n)}$. For the exceptional n, $p^2 \mid n(N-n)$ for some $p \ge z$, and $\sum_{p\ge z} \sum_{p^2 \mid n(N-n)} w(n) \ll N^{\varepsilon} \sum_{p\ge z} N/p^2 \ll N^{1-\alpha+\varepsilon}$. □

Lemma 4.2 (Prime factors)**.** *If n is as in Lemma 4.1, $n(N-n)$ square-free and $w_n > 0$, then $\Omega(n(N-n)) < 1/\lambda + 2/\beta_1$ and $\max(\Omega(n), \Omega(N-n)) < 1/\lambda + 1/\beta_1$.*

Proof. As in [3, Lemma 3.2]: write $w_n = 1 - \lambda(T_1 + T_2)$ with T_1, T_2 the sums over the primes dividing n, resp. $N-n$. Each term is non-negative, and since $n < N = y_1^{1/\beta_1}$ has fewer than $1/\beta_1$ prime factors $\ge y_1$, $T_1 \ge \Omega(n) - 1/\beta_1$; similarly for T_2. Then $w_n > 0$ gives $T_1 + T_2 < 1/\lambda$, hence both claims. □

Proposition 4.3 (Main term)**.** *Let $\theta = 2/5$, $\theta_A \in (0, \theta)$, $s_A = \theta_A/\alpha$, $s_B = (\theta - \theta_A)/\alpha$, and let F_1, f_1 be the upper and lower functions of the linear sieve. Put*

$$L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B), \qquad R = 2\int_{\alpha}^{\beta_1} \Big(1 - \frac{u}{\beta_1}\Big) U(\theta - u)\, \frac{\mathrm{d}u}{u},$$

where $U(t) = \min_{t_A + t_B = t} F_1(t_A/\alpha)F_1(t_B/\alpha)$. Then

$$W \;\ge\; (L - \lambda R + o(1))\, X\, V, \qquad V = \prod_{\substack{2<p<z \\ p\nmid N}} (1 - h(p))^2,$$

and

$$XV = \big(1 + o(1)\big)\, 8\, \mathfrak{S}_r(N)\, c_h^2\, \frac{N}{\alpha^2 (\log N)^2}, \qquad c_h = \lim_{z\to\infty} (\log z) \prod_{2<p<z} (1 - h(p)) = 1.50136\ldots \tag{15}$$

Proof. Apply the vector sieve inequality $ab \ge a^- b^+ + a^+ b^- - a^+ b^+$ [2] with $a = 1_{(n,P)=1}$, $b = 1_{(N-n,P)=1}$ and the Rosser–Iwaniec linear sieve weights [7] of levels N^{θ_A} and $N^{\theta-\theta_A-\varepsilon}$. By (14) the sequence $\{n \in \mathcal{A} :\ d \mid n\}$ with the weight w has density $h(d)$, with $h(p) = (2p-1)/(p(p+1))$ on the split primes and $h(q) \ll q^{-2}$ on the inert ones; since $\sum_{p<z,\, p\equiv 1\,(4)} 1/p = \frac{1}{2}\log\log z + O(1)$, each side is a sieve of dimension 1, i.e. $\prod_{w\le p<z,\, p\nmid N}(1-h(p))^{-1} \le (\log z/\log w)(1 + K/\log w)$ with an absolute K. The combined moduli are $\le N^{\theta-\varepsilon}$, and Theorem 3.2 shows that the remainders contribute $O(N^{1-\varepsilon/3}) = o(XV)$. This gives $\sum_{(n(N-n),P)=1} w(n) \ge (L + o(1))XV$. For the weights, the subsequence $\{n \in \mathcal{A} :\ p \mid n\}$ with $z \le p < y_1$, $p \nmid N$, has density $h(p)h(d)$ for $d \mid n$

and the same structure, and the upper bound vector sieve at total level $N^{\theta-u-\varepsilon}$, $p=N^u$, bounds its sifted weight by $(U(\theta-u)+o(1))h(p)XV$; the remainders are again controlled by Theorem 3.2 since $d_3(pde)\ge d(de)$ absorbs the sum over p. Since $\sum_{p\equiv 1\,(4),\ N^u\le p<N^{u+\mathrm{d}u}} h(p)=(1+o(1))\,\mathrm{d}u/u$ by the prime number theorem in progressions ($h(p)=2/p+O(p^{-2})$ on half of the primes), and the side $N-n$ contributes the same amount, the weighted sum is bounded by $(R+o(1))XV$ with the factor 2 in R accounting for the two sides. Finally, by (13), Lemma 3.1 and (1),

$$XV=8N\prod_{p\mid N^\flat}\frac{\beta_p^*}{1-p^{-2}}\prod_{\substack{2<p<z\\ p\nmid N}}(1-h(p))^2=8N\,\mathfrak{S}_r(N)\prod_{2<p<z}(1-h(p))^2\prod_{\substack{p\mid N\\ p\ge z}}(1-h(p))^2,$$

and the last product is $1+O(\omega(N)/z)$. The constant c_h exists by Mertens' theorem in arithmetic progressions: $\prod_{2<p<z}(1-h(p))=P_1(z)^2\prod_{2<p<z}(1-h(p))(1-1/p)^{-2\cdot 1_{p\equiv 1(4)}}$ with $P_1(z)=\prod_{p<z,\,p\equiv 1(4)}(1-1/p)\sim(2e^{-\gamma}/\kappa)^{1/2}(\log z)^{-1/2}$, $\kappa=\frac{\pi}{4}\prod_{q\equiv 3(4)}(1-q^{-2})$, the last product converging absolutely. Numerically $\kappa=0.672386\ldots$, the correction product is $0.898991\ldots$ and $c_h\in[1.50136,1.50137]$ (script `singular_constant.py`, primes up to 10^7 with explicit tail bounds). □

α	λ	β_1	$1/\lambda+2/\beta_1$	$1/\lambda+1/\beta_1$	$L\ge$	$R\le$	$L-\lambda R\ge$
9/250	3/10	3/10	10	20/3	0.998974	2.816690	0.153967
4/125	7/30	7/20	10	50/7	0.999637	3.471422	0.189639
9/250	1/3	3/10	29/3	19/3	0.998974	2.816690	0.060078

TABLE 1. Certified bounds at level $\theta=2/5$ with $\theta_A=\theta_B=1/5$. The first line gives $\Omega(n(N-n))\le 9$ and $\max(\Omega(n),\Omega(N-n))\le 6$; the second $\Omega(n(N-n))\le 9$ with the larger constant; the third the same bounds as the first with $\lambda=1/3$.

4.1. **Numerical verification.**

Lemma 4.4. *The bounds of Table 1 hold.*

Method of verification. The computation is carried out by the script `certify_modular.py`, which follows [3, Lemma 4.1]. The functions F_1, f_1 are evaluated from their closed forms $F_1(s)=2e^\gamma/s$ on $(0,3]$ and $f_1(s)=2e^\gamma\log(s-1)/s$ on $[2,4]$ in interval arithmetic (`mpmath.iv`, 80 bits) and continued by the delay-differential equations $(sF_1)'=f_1(s-1)$, $(sf_1)'=F_1(s-1)$ on the grid $10^{-3}\mathbb{Z}$ with outward rounding, using that F_1 decreases and f_1 increases. L is bounded below through $L=f_1(s_A)F_1(s_B)+F_1(s_A)(f_1(s_B)-F_1(s_B))$ with $f_1-F_1\le 0$. R is bounded above by a Riemann sum over 4000 cells: on a cell $[u_a,u_b]$ the factor $(1-u/\beta_1)/u$ is largest at u_a, the level $\theta-u$ is smallest at u_b, and U is bounded by the product $F_1(t/\alpha)F_1((\theta-u_b-t)/\alpha)$ at one admissible t. The rational numbers $1/\lambda+2/\beta_1$ and $1/\lambda+1/\beta_1$ are computed exactly. □

Proof of Theorem 1.1. Take the first line of Table 1. By Proposition 4.3 and Lemma 4.4, $W\ge(0.153967+o(1))XV$. Since $w_n\le 1$ and $w(n)\ge 0$, the sum of $w(n)$ over the n with $(n(N-n),P)=1$ and $w_n>0$ is at least W; removing the exceptional n of Lemma 4.1 costs $O(N^{1-\alpha+\varepsilon})$. By Lemmas 4.1 and 4.2 every remaining n has n, $N-n$ sums of two squares, $\Omega(n(N-n))\le 9$, $\max(\Omega(n),\Omega(N-n))\le 6$ and $w(n)=16\cdot 2^{\Omega(n(N-n))}\le 2^{13}$. Hence the number of such n is at least

$$\big(0.153967+o(1)\big)\,2^{-13}\cdot 8\,\mathfrak{S}_r(N)\,c_h^2\,\frac{N}{\alpha^2(\log N)^2}\ \ge\ 0.26\,\mathfrak{S}_r(N)\frac{N}{(\log N)^2}$$

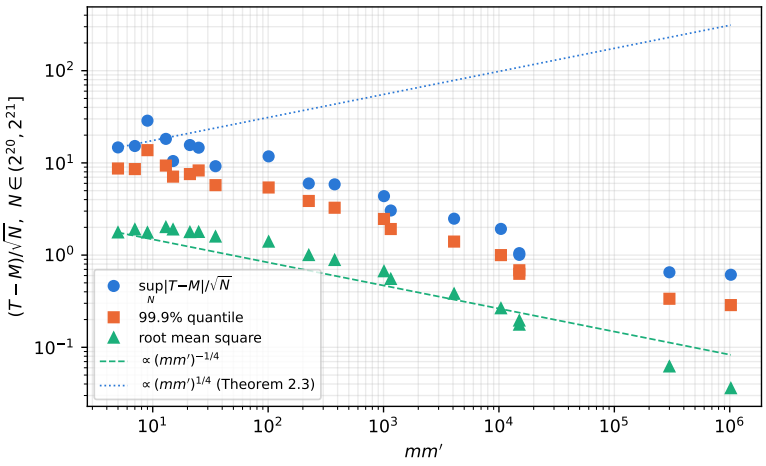


FIGURE 1. The data of Table 2 for $X = 2^{21}$ (all pairs computed), with the lines $(mm')^{-1/4}$ and $(mm')^{1/4}$ for comparison. The supremum does not grow with the level; the root mean square decreases like $(mm')^{-1/4}$.

for large N, since $0.153967 \cdot 2^{-10} \cdot c_h^2 \cdot (250/9)^2 \geq 0.2615$. Finally $\rho_p > 1$ for all odd p: for $p \equiv 1 \pmod 4$, $p(p+1)(p-1)^2 - (p^2-p+1)^2 = p^3 - 4p^2 + 3p - 1 = p^2(p-4) + 3p - 1 > 0$ for $p \geq 5$ (e.g. $\rho_5 = 480/441$), and for $q \equiv 3 \pmod 4$, $\rho_q > (q+1)/q > 1$. □

Remark 4.5. If the error term in Theorem 2.3 were $O(N^{1/2+\varepsilon})$ uniformly in m, m' (Conjecture 5.1), Theorem 3.2 would hold with $D = N^{1/2-\varepsilon}$, and the same computation at $\theta = 1/2$ with $\alpha = 6/125$, $\lambda = 1/3$, $\beta_1 = 17/50$ (so $1/\lambda + 2/\beta_1 = 151/17 < 9$, $1/\lambda + 1/\beta_1 = 101/17 < 6$) certifies $L \geq 0.997718$, $R \leq 2.498893$, $L - \lambda R \geq 0.164754$, hence $\Omega(n(N-n)) \leq 8$ and $\max(\Omega(n), \Omega(N-n)) \leq 5$ (script `certify_modular.py` with the argument `1/2`).

5. Numerical evidence and a conjecture

The quantities $T(m, m'; N)$ for all $N \leq X$ are obtained at once by a fast Fourier transform convolution of the sequences $r(a)1_{m|\cdot}$ and $r(b)1_{m'|\cdot}$, and $M(m, m'; N)$ from (5)–(7) with the 2-adic density computed exactly modulo 2^9 (script `sup_error_scan.py`). Table 2 and Figure 1 report the statistics of $(T-M)/\sqrt{N}$ over all $N \in (X/2, X]$ with $(N, mm') = 1$ and $v_2(N) \leq 3$, for $X = 2^{21}$ and $X = 2^{23}$. Three consistency checks: for $mm' \in \{1, 3\}$ (levels 4 and 12, where $S_2(\Gamma_0(L)) = 0$) the error is identically zero, so the main term (5) is exact; for $(m, m') = (1, 5)$ and $(9, 1)$ the error vanishes identically for $N \equiv 2 \pmod 4$, because the newforms of levels 20 and 36 have $a(2) = 0$ and there are no oldforms with $a(2) \neq 0$; and the density (7) was checked against exact counts modulo p^2 for $p \leq 13$.

Two features stand out. First, the root mean square decreases like $(mm')^{-1/4}$. By the Rankin–Selberg method, the mean square of $a_G(n)/\sqrt{n}$ over $n \leq Y$ tends to $(4\pi)^2\|G\|^2/\mathrm{vol}(\Gamma_0(4mm')\backslash\mathbb{H})$, and the volume is $\asymp mm'$; so the data say $\|G\| \asymp (mm')^{1/4}$, i.e. Lemma 2.1 is sharp up to N^{ε}. (Waibel's lower bound [12, Theorem 3] gives only $\|G\|^2 \gg 1/\min(m, m')$ for our forms.) Second, the supremum over N does not grow with mm' at all—it decreases—while Theorem 2.3 allows growth like $(mm')^{1/4}$. The gap is the Cauchy–Schwarz step in Lemma 2.2: the coefficient $a_G(N)$ is a sum over $\asymp mm'$ Hecke eigenforms of terms whose moduli are controlled by $\|G\|$, and the inequality is sharp only if their phases align, which the data say they never do. The growth in N

(m,m')	mm'	rms	99.9% quantile	sup, $X=2^{21}$	sup, $X=2^{23}$
$(1,5)$	5	1.76	8.7	14.7	17.3
$(9,1)$	9	1.76	13.7	28.7	33.8
$(3,5)$	15	1.90	7.1	10.5	12.2
$(1,101)$	101	1.40	5.4	11.8	12.1
$(13,29)$	377	0.89	3.3	5.9	6.8
$(1,1009)$	1009	0.67	2.5	4.4	5.3
$(61,67)$	4087	0.38	1.4	2.5	2.4
$(101,103)$	10403	0.27	1.0	1.9	1.8
$(1,15015)$	15015	0.18	0.6	1.0	–
$(1009,1013)$	1022117	0.036	0.29	0.61	0.55

TABLE 2. Statistics of $(T(m,m';N)-M(m,m';N))/\sqrt{N}$ over all admissible $N\in(X/2,X]$ ($\approx 10^6$ values for $X=2^{21}$, $4\cdot 10^6$ for $X=2^{23}$; the run with $X=2^{23}$ covers a selection of the pairs).

between $X=2^{21}$ and 2^{23} is by a factor 1.0–1.2, as expected from Deligne's bound (the suprema are attained at N with 16–96 divisors). We therefore expect:

Conjecture 5.1. For coprime odd m,m' and $N\geq 1$ with $(N,mm')=1$ and $v_2(N)\leq 1$, $T(m,m';N)=M(m,m';N)+O_\varepsilon\big(N^{1/2+\varepsilon}\big)$ uniformly in m,m'. Consequently $r(n)r(N-n)$ has level of distribution $1/2$, and the bounds of Theorem 1.1 can be replaced by 8 and 5.

The conjecture is the exact analogue, for $r(n)$ and the sum orientation, of the uniformity in Topacogullari's theorem [11] for the divisor function, without the $7/64$; it would follow from a uniform bound for the quantity $\sum_f\langle\Theta,f\rangle a_f(N)/\|f\|^2$ over a Hecke basis, i.e. from a first moment of the central values $L(1/2,f)L(1/2,f\otimes\chi)$ twisted by $\lambda_f(N)$ over the weight-2 forms of level $4mm'$.

DATA AND CODE AVAILABILITY

The scripts behind Table 2 and Figure 1 (`sup_error_scan.py`), Table 1 (`certify_modular.py`), the choice of the sieve parameters (`modular_sieve_scan.py`) and the constant c_h (`singular_constant.py`), together with their outputs, are available at

https://github.com/Ruqing1963/two-squares-theta-level

and archived at Zenodo, doi:10.5281/zenodo.23438288.

During the preparation of this work the author used the Claude language model and the `mpmath` Python library for the computer-assisted interval arithmetic verification of Section 4 and for the numerical experiments of Section 5. All analytical derivations and proofs were independently audited and verified by the author.

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